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# Verification on out-of-distribution detectors under natural perturbations

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**Abstract**

Out-of-distribution (OOD) detectors play a vital role in distinguishing between OOD data and in-distribution data. However, the vulnerability of OOD detectors to natural perturbations, such as rotation and lighting variations, can potentially lead to catastrophic accidents in safety-critical applications. Current attack techniques lack robustness guarantees for OOD detectors. Neural network (NN) verification methods are limited to standard NN structures and cannot be applied to OOD detectors due to their non-standard structure. To address this issue, we propose a verification framework called *Vood* that offers robustness guarantees for OOD detectors under natural perturbations. Our approach begins by proving the Lipschitz continuity of most OOD detection functions under natural transformations. We then estimate the Lipschitz constant using Extreme Value Theory, incorporating a dynamically estimated safety factor. *Vood* transforms the verification problem into an optimization challenge, which is then effectively addressed using space-filling Lipschitz optimization techniques. Additionally, *Vood* is a *black-box* verifier, which can tackle natural perturbations on a wide range of OOD detectors. Through empirical analysis, we demonstrate that *Vood* outperforms baseline methods in both accuracy and efficiency. Our work represents a pioneering effort in establishing robustness verification for OOD detectors with provable guarantees.

**Keywords** OOD detector · Natural perturbation verification · Lipschitz optimization

## 1 Introduction

Neural networks (NNs) have demonstrated outstanding performance across various domains, including medical diagnosis (Sarvamangala & Kulkarni, 2022; Abdou, 2022) and autonomous driving (Chen et al., 2021; Kocić et al., 2019). Most research focuses on closed-world machine learning, assuming that training and testing data share the same distribution. In real-world scenarios, NNs often face an open-world scenario, where data distributions differ. Ideally, networks should avoid making predictions on OOD inputs, but

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they are susceptible to confidently predicting adversarial OOD examples. Nguyen et al. (2015). For instance, if a bird image is given as input to a road sign recognition neural network, the network might wrongly make confident predictions. Such behavior can lead to catastrophic consequences in safety-critical applications such as autonomous driving, where anomalous data must be handled differently. To mitigate this issue, various OOD detectors (Liu et al., 2020; Wang et al., 2022; Lee et al., 2018) have been developed to identify OOD data before they are presented to a neural network.

In an open-world machine learning scenario, the initial step involves processing the input data $x$ through an OOD detector denoted as $f$ as demonstrated in Fig. 1. This detector maps the input $x$ to a score $f(x) \in \mathbb{R}$ and compares it with a threshold $\tau$. When $f(x) > \tau$, the input is classified as OOD; otherwise, it is considered ID data. Methods for designing the score function $f$ can be broadly categorized into three groups: classification-based (Hendrycks & Gimpel, 2016; Liang et al., 2017; Hendrycks et al., 2022), density-based (Liu et al., 2020), and distance-based approaches (Lee et al., 2018).

In spite of the promising results of the mentioned OOD detectors, they face significant robustness challenges. As depicted in Fig. 1, even a specialized OOD detector for road signs can erroneously classify a stop sign as OOD during rainy days. This can cause severe safety issues, since OOD detectors are mostly applied in practical open world situations such as autonomous driving systems, where natural perturbations occur all the time.

To address this challenge, researchers have employed adversarial attacks and learning methods to enhance the robustness of OOD detectors. Notable examples include ALOE (Chen et al., 2020), ATOM (Chen et al., 2021), and ATD (Azizmalayeri et al., 2022). Despite these advanced techniques, none provide a robustness guarantee. This lack of assurance is particularly concerning for safety-critical applications, underscoring the urgent need to establish the robustness of OOD detectors before real-world deployment.

Existing verification techniques (Singh et al., 2019; Müller et al., 2021; Fazlyab et al., 2020; Xue et al., 2022) cannot be directly applied to OOD detectors because these techniques are specifically designed for standard NNs, utilizing a layer-by-layer analysis approach. In contrast, OOD detectors involve complex functions that modify NN results or may not include a neural network structure at all. Furthermore, most NN verification tools require detailed knowledge of the NN's architecture, which is often inaccessible

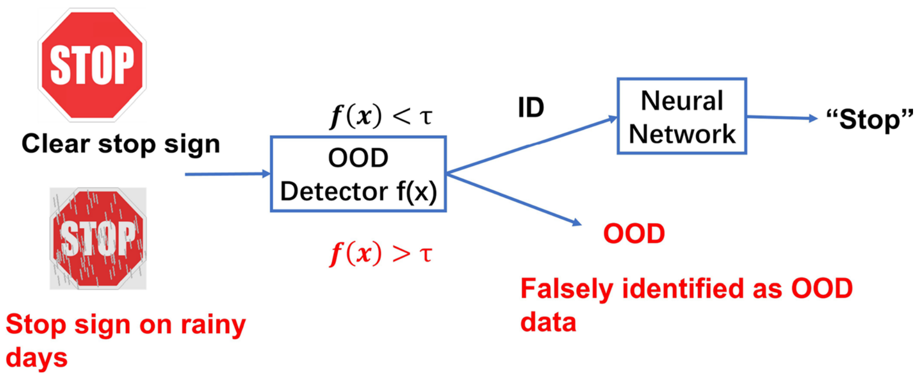



**Fig. 1** Working principles of OOD detectors in an open-world machine learning system. A sample with $f < \tau$ will be passed to the NN. A stop sign can be misclassified as OOD data on rainy days. Natural perturbations can cause misjudgment of OOD detectors

in practical applications of OOD detectors. Additionally, there is limited discussion on naturally occurring perturbations, which are evidently prevalent in real-world scenarios.

To address this concern, we introduce a verification framework, *Vood*, designed to provide a safety guarantee for the robustness of OOD detectors against natural perturbations. By specifying a range for perturbation parameters, such as rotation angles, *Vood* assesses the output range of the OOD score function. Subsequently, it compares this output range with a predefined threshold $\tau$ to ascertain the safety of the OOD detector, as illustrated in Fig. 2. In Fig. 2, distinct input ranges $[a_1, b_1]$, $[a_2, b_2]$, and $[a_3, b_3]$ for the perturbation parameter $\delta$ are provided. *Vood* calculates the output range (blue bars) of the score function and compares it with the predefined threshold $\tau$. When the output range intersects with the threshold, the perturbation is deemed unsafe. *Vood* focuses on natural perturbations described by a limited set of parameters, handling a broad range from rotations to hue, saturation, and lightness (HSL) variations. Certain complex weather changes such as rain, fog, snow, and sun flare can also be effectively modulated with a limited number of parameters (Buslaev et al., 2020; Tremblay et al., 2021; Jung et al., 2020) and further analyzed using *Vood*.

As far as we know, *Vood* is the first verifier designed for analyzing the robustness of OOD detectors under natural perturbations, providing a theoretical robustness guarantee. The main contribution of our work is summarized as follows:

- *Vood* is the first verification approach analyzing OOD detectors under natural perturbations, including rotation or changes in HSL values. *Vood* solves the verification problem by calculating the reachability of OOD score functions, providing a safety guarantee.
- *Vood* addresses the reachability problem through global optimization, employing a space-filling scheme. We demonstrate that the majority of out-of-distribution (OOD) score functions exhibit Lipschitz continuity under natural perturbations. We estimate the Lipschitz constant with Extreme Value Theory (EVT) and then amplify it using a dynamically estimated safety parameter for conservative overestimation. The space-filling approach ensures efficient convergence in Lipschitz optimization, even with a comparatively large Lipschitz constant.
- *Vood* stands as a pioneering approach for assessing the robustness of OOD detectors under natural perturbations. *Vood* effectively handles diverse OOD detectors with non-

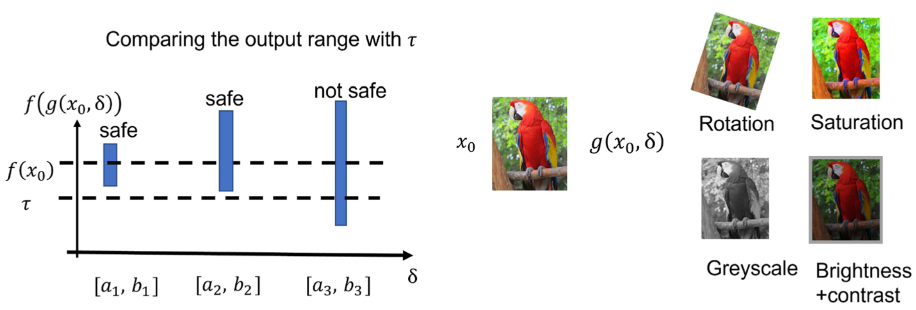



**Fig. 2** Overview of *Vood* handling. The natural perturbed samples (blue points) are generated by function $g$ of original $x_0$ and perturbed parameter $\delta$. $\delta$ can be a rotation angle, saturation grades, etc. The output ranges (blue bars) are calculated corresponding to the ranges of $\delta$. The perturbation range $[a_3, b_3]$ is not safe since its output range meets the threshold $\tau$ (Color figure online)

standard neural network structures, overcoming limitations in existing verification tools. *Vood* excels in managing OOD detectors in complex black-box systems, enhancing its practical utility in real-world scenarios.

## 2 Related work

### 2.1 Out-of-distribution detection

The classification-based methods generate the OOD score function from the neural network outputs. MaxSoft (Hendrycks & Gimpel, 2016) uses maximal softmax output. Openmax (Bendale & Boult, 2016) replaces softmax with an openmax layer and calibrates logits using EVT model. G-OpenMax (Ge et al., 2017) enhances OpenMax with GANs. ODIN (Liang et al., 2017) separates OOD and ID data with temperature scaling and input perturbations. MaxLogit (Hendrycks et al., 2022) uses maximum logit, and introduces KL-divergence-based matching (Hendrycks et al., 2022). Vim (Wang et al., 2022) proposes virtual-logit matching. Density-based approaches utilize probabilistic models to represent the ID data and identify samples as OOD when they fall into low-density regions. Representative methods includes energy-based model (Liu et al., 2020) and its extension to GNNs (Wu et al., 2023), and Normalizing flow (Kobyzev et al., 2020). Distance-based methods such as (Lee et al., 2018) uses Mahalanobis distance for distinguishing between ID and OOD samples, and has shown promising results when it combines with transformers (Podolskiy et al., 2021). Large language models (LLMs) are integrated into OOD detection systems in Cao et al. (2024).

### 2.2 Robustness of OOD detection

In addition to accuracy, there's a growing interest in enhancing the robustness of OOD detection. ACET (Hein et al., 2019) discovers that ReLU models struggle in OOD detection under perturbations, even with temperature rescaling. Consequently, they proposed adversarial enhancement training for improved robustness. In Meinke and Hein (2020) a Bayesian framework is employed for modelling ID and OOD data separately. NN is trained through a maximum likelihood estimator, while it is limited to ReLU models. Good (Bitterwolf et al., 2020) utilizes interval bound propagation (IBP) to set upper bounds on $l_\infty$ -norm confidence, minimizing them for accuracy and robustness. However, IBP's bounds are often loose and limited to $l_\infty$-norm. ATOM (Chen et al., 2021) improve robustness via adversarial training and selective use of auxiliary outlier data, creating a tight ID-OOD boundary. These methods strengthen specific OOD detectors and test them experimentally, but most of them are limited to $l_p$-norm attacks. In Yoon et al. (2022), the Evaluation-via-Generation (EvG) method is introduced as an attack strategy against OOD detectors. Wang et al. (2022) proposes a watermarking framework to improve OOD detectors by embedding watermarks into input data. Although both methods address functional perturbations, they lack theoretical analysis of general detector robustness under natural perturbations.

### 2.3 Verification

Several neural network verification techniques have been developed to provide a reliable safety guarantee (Liu et al., 2021). Abstract interpretation (Singh et al., 2019; Müller et al., 2021) uses domains like boxes, zonotopes, or polyhedra to approximate inputs, propagating results through the network. Constraint solver-based methods (Fazlyab et al., 2020; Xue et al., 2022) encodes the targeted NN, then tackles the verification by various solvers including linear or mixed-integer linear programming. To deal with the nonlinear property of NN, some researchers (Shi et al., 2020) approximate nonlinear layers linearly, demanding detailed NN structure knowledge. Ruan et al. (2018) optimizes reachability with limited perturbation dimensions. However, those verification methods are specifically designed for neural network classifiers, focusing on $l_p$ norm perturbation, direct applications to OOD detectors may prove unsuitable or yield substantial estimation errors, due to the non-standard NN structure of score functions. Moreover, since OOD detectors normally are applied to open-world scenarios, verifiers for OOD detectors should be able to tackle natural perturbations such as rotations and lighting variations.

## 3 Problem formulation

**Definition 1** (*OOD detector*) Given an input sample $x$, an OOD detector with threshold $\tau$ provides an OOD score function $f(x)$. For function value $f(x) > \tau$, input $x$ will be defined as OOD data. For $f(x) < \tau$, $x$ will be regarded as in-distribution data.

The score function $f(x)$ is a complex function that may incorporate a modified neural network structure, depending on the category of OOD detectors. For a well-trained OOD detector, we adopt the score of the true positive rate (TPR) at 95% as the threshold $\tau$.

**Definition 2** (*Natural perturbation on OOD detectors*) Given a detector $f(x)$, an input $x_0 = (x_0^1, x_0^2, ..., x_0^k)$ and a threshold $\tau$, a natural perturbation $g$ with respect to parameter $\delta = (\delta^1, \delta^2, ..., \delta^m)$ is defined as $g(x_0, \delta) = (y^1, y^2, ..., y^k)$. The corresponding perturbed score function is $f(g(x_0, \delta)) = f(y^1, y^2, ..., y^k)$.

The natural perturbation $g(x_0, \delta)$ characterizes the functional transformation induced by natural or environmental changes. Here, $\delta$ serves as the associated parameter, reflecting the specific nature of the transformation. For instance, if $g(x_0, \delta)$ represents rotation, $\delta$ corresponds to the rotation angle. In the case of alterations in hue, saturation, and lightness, $\delta = (h, s, l)$ constitutes a three-dimensional parameter quantifying the extent of the transformation.

**Definition 3** (*Successful natural attack on OOD detectors*) Given a detector $f(x)$, an input $x_0 = (x_0^1, x_0^2, ..., x_0^k)$ and a threshold $\tau$, a bounded natural attack $A$ creates perturbed input set $Y = \{g(x_0, \delta), \delta \in [a, b]^m\}$. $A$ is a successful attack, if a $y' \in Y$ exists, where $\mathrm{sgn}[f(x_0) - \tau] \neq \mathrm{sgn}[f(y') - \tau]$. $[a, b]^m$ is the perturbation range of the natural attack.

A successful natural attack involves a natural perturbation that induces a misclassification of the OOD detector.

**Definition 4** (*Safety of OOD detectors*) Given an OOD detector $f(x)$ an input $x_0$, a threshold $\tau$, the detector is safe with respect to natural perturbation $g(x, \delta)$ and perturbation range $[a, b]^m$, if

$$\forall y \in Y : \mathrm{sgn}[f(x_0) - \tau] = \mathrm{sgn}[f(y) - \tau] \tag{1}$$

with $Y = \{g(x_0, \delta), \delta \in [a, b]^m\}$.

**Definition 5** (*Reachablity of OOD detector*) Given a detector $f(x)$, a functional perturbation $g(x_0, \delta)$ at sample $x_0$, and a perturbation range $\delta \in [a, b]^m$, $[l, u]$ is reachable with respect to $f$, $g$, $x_0$, $[a, b]^m$ and tolerance $\zeta$ if $\forall \delta \in [a, b]^m$, the inequality

$$l - \zeta \le \min f(g(x_0, \delta)) \le l + \zeta$$

$$\textit{and } u - \zeta \le \max f(g(x_0, \delta)) \le u + \zeta$$

always hold.

We can further determine the safety of the detector based on the tolerance $\zeta$ and the reachable range $[l, u]$. For ID sample with $f(x_0) < \tau$, the perturbation $g(x, \delta)$ with $\delta \in [a, b]^m$ is safe if $u + \zeta < \tau$. For OOD sample with $f(x_0) > \tau$, the perturbation $g(x, \delta)$ with $\delta \in [a, b]^m$ is safe if $l - \zeta > \tau$.

## 4 Lipschitz continuity of OOD score function

We establish the Lipschitz continuity of the perturbed score function $f(g(x_0, \delta))$ concerning the natural perturbation parameter $\delta$. We achieve this by demonstrating the Lipschitz continuity of both $f(x)$, $g(x_0, \delta)$, and the chain rule (Ambrosio & Dal Maso, 1990). We begin by revisiting the definition of Lipschitz continuity.

**Theorem 1** (*Lipschitz Continuity* (Armijo, 1966)) *Given two metric spaces* $(X, d_X)$ *and* $(Y, d_Y)$, *where* $d_X$ *and* $d_Y$ *are the metrics on the sets* $X$ *and* $Y$ *respectively, a function* $f : X \to Y$ *is called Lipschitz continuous if there exists a real constant* $K \ge 0$ *such that, for all* $x_1, x_2 \in X$: $d_Y(f(x_1), f(x_2)) \le K d_X(x_1, x_2)$. *K is called the Lipschitz constant of f. The smallest K is called the Best Lipschitz constant, denoted as* $K_{best}$.

We further have the following lemma.

**Lemma 1** *Let* $f : \mathbb{R}^n \to \mathbb{R}^m$, *if* $|\partial f(x)/\partial x| \le K$ *for all* $x \in [a, b]^n$, *then f is Lipschitz continuous on* $[a, b]^n$ *and K is its Lipschitz constant, where* $| * |$ *represents a norm operator.*

A group of OOD score functions denoted as $f(x)$, is formulated by leveraging NN output. Considering that the majority of neural networks exhibit Lipschitz continuity (Virmaux & Scaman, 2018; Ruan et al., 2018), the logit layer output associated with each label, denoted as $l_j(x)$, maintains Lipschitz continuity concerning the input variable $x$. By capitalizing on the Lipschitz continuity inherent in most neural networks and the devised construction of the OOD detector, we establish the following theorem.

**Theorem 2** *The OOD score function, denoted as f(x), is considered Lipschitz continuous with a Lipschitz constant L. This property holds true when f(x) is formulated through adjustments to neural network outputs, as observed in the case of OOD detectors utilizing classification methods and density-based approaches.*

***Proof*** Incorporating temperature scaling, as observed in the case of a MaxSoftmax OOD detector, yields the following score function:

$$f(x) = -\max_j S_j(x), S_j(x) = \frac{e^{l_j(x)/T}}{\sum_{i=1}^{N} e^{l_i(x)/T}} \tag{2}$$

We estimate the partial derivative:

$$\left|\frac{\partial S_j(x)}{\partial l_i(x)}\right| = \begin{cases} |S_j(x)(1 - S_j(x))/T| \leq 1/T, & i = j \\ |-S_j(x)S_i(x)/T| \leq 1/T, & i \neq j \end{cases} \tag{3}$$

Thus $|\partial f(x)/\partial x| \leq 1/T$. The score function is Lipschitz continuous.

For a density-based method such as an energy-based OOD detector, we have the following score function:

$$f(x) = -T \log \sum_j e^{l_j(x)/T} \tag{4}$$

We further calculate:

$$\left|\frac{\partial f(x)}{\partial l_i(x)}\right| = e^{l_j(x)/T} / \sum_j e^{l_j(x)/T} \leq 1 \tag{5}$$

Since $l_i(x)$ of most NN is Lipschitz continuous (Virmaux & Scaman, 2018) and chain rule we have proved that $f(x)$ is also Lipschitz continuous on $x$. □

We further discuss the Lipschitz continuity of $g(x_0, \delta)$ on $\delta$. We discuss the situations of $\delta$ representing changes in the aspect of color and rotation separately.

**Theorem 3** *The natural perturbation on input $x_0$, donated as $g(x_0, \delta)$, is Lipschitz continuous on $\delta$. When the transformation is caused by changes in brightness, contrast, sharpness, color, saturation, or rotation.*

***Proof*** Changing of brightness, contrast, sharpness, color, or saturation are realized by interpolation and extrapolation between two images as shown in Eq. (6)

$$g(x_0, \delta) = x_{de} \cdot (1.0 - \delta) + x_0 \cdot \delta \tag{6}$$

The perturbed input $g(x_0, \delta)$ is created by a linear combination of the original image $x_0$ and a degenerated image $x_{de}$. $x_{de}$ is fixed for a given $x_0$ and a predefined perturbation type. It has the same size as the original $x_0$ and for different perturbation types, it can be black, greyscale of $x_0$, etc. We compute the corresponding partial derivative:

$$|\partial g(x_0, \delta)/\partial \delta| = |x_0 - x_{de}| \tag{7}$$

which is a constant for a predefined sample and perturbation type. Thus $g(x_0, \delta)$ is Lipschitz continous on $\delta$.

Image rotation is realized by matrix multiplication.

$$g(x_0, \delta) = A \cdot B \cdot C \cdot x_0 \tag{8}$$

A,B and C are

$$A = \begin{bmatrix} 1 & 0 & c_x \\ 0 & 1 & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} \cos\delta & \sin\delta & 0 \\ -\sin\delta & \cos\delta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & -c_x \\ 0 & 1 & -c_y \\ 0 & 0 & 1 \end{bmatrix} \tag{9}$$

where $c_x, c_y$ is the coordinate of the rotation center and $\delta$ is the rotation angle.

$$|\partial g/\partial\delta| \leq |A||C||x_0| \cdot \begin{vmatrix} -\sin\delta & \cos\delta & 0 \\ -cos\delta & -sin\delta & 0 \\ 0 & 0 & 1 \end{vmatrix} \leq 5|A||C||x_0| \tag{10}$$

For a perturbation of image rotation around a predefined point $c$, the value of $A, B, x_0$ is fixed. Thus $|\partial g/\partial\delta|$ is bounded and $g$ is Lipschitz continuous on $\delta$. □

## 5 Preliminary estimation of Lipschitz constant

*Vood* first calcuates the Lipschitz constant $L_e$ by estimating maximal gradient norm within the range $\delta \in [a, b]$. When sampling $\partial f/\partial\delta$ in $\delta \in [a, b]$ randomly, the $|\partial f/\partial\delta|$ can be treated as independent and identically distributed random variables with an upper bound. Their distribution follows the Fisher-Tippett-Gnedenko Theorem of EVT.

**Theorem 4** (*Fisher-Tippett-Gnedenko Theorem*) *Let $Z_1, Z_2 ... Z_n$ be a sequence of independent and identically distributed random variables with cumulative distribution function (CDF) F, and the CDF of* $\max\{Z_1, ..., Z_n\}$ *denoted as $F_n(z)$. Suppose that there exist two sequences of real numbers $c_n > 0$ and $d_n \in R$ such that the following limits converge to a non-degenerate distribution function*: $\lim_{z\to\infty} F_n(c_n z + d_n) = G(z)$. *$G(z)$ will converge to one of three possible limiting distributions, which are characterized by three types*, 1. *Reverse Weibull distribution*, 2. *Gumbel distribution*, 3. *Fréchet distribution*:

$$1.\ G(z) = \begin{cases} \exp\{-(-(\frac{z-d}{c}))^{\alpha}\} & z \leq d \\ 1 & z > d \end{cases} \tag{11}$$

$$2.\ G(z) = \exp\{-\exp(-(\frac{z-d}{c}))\} \tag{12}$$

$$3.\ G(z) = \begin{cases} 0 & z \le d \\ \exp\{-(\frac{z-d}{c})^{-\alpha}\} & z > d \end{cases} \tag{13}$$

Since the $\delta \in [a, b]$ is a bounded range, the sampled $\partial f / \partial \delta$ is bounded, with the distribution belonging to the reverse Weibull family. We sample $M$ gradients in the range $[a, b]$ and repeat it $N$ times to fit the reverse Weibull Distribution. $\partial f / \partial \delta$ is calculated in a numerical way. The process for estimating the Lipschitz constant is in Algorithm 1.

**Algorithm 1** Lipschitz constant estimation

**Input:** $M, N, a, b, f(x), x_0, \tau, g$
**Output:** $L_e$
1: **for** $k \leftarrow 1$ to $N$ **do**
2: Generate random $\delta^1, \delta^2, ...\delta^M$ in $[a, b]$
3: Calculate $|\partial f/\partial \delta^1|, |\partial f/\partial \delta^2|, ...|\partial f/\partial \delta^M|$
4: $l_k = \max\{|\partial f/\partial \delta^1|, |\partial f/\partial \delta^2|, ...|\partial f/\partial \delta^M|\}$
5: **end for**
6: Fit $l_1, ..., l_N$ to the reverse Weibull distribution, $L_e \leftarrow b$

## 6 Discussion

The Lipschitz estimation method via sampling and EVT has been applied in Weng et al. (2018); Hein and Andriushchenko (2017). In Goodfellow (2018) the author has also argued that the method can not be applied to gradient masking, i.e. perturbation involving step function. However, we here deal with more natural perturbation caused by environment or weather change, in which a step function is not included. In contrast to Weng et al. (2018) and Hein and Andriushchenko (2017), *Vood* distinguishes itself by not mandating an exact Lipschitz constant estimation. Instead, we initiate an approximate Lipschitz constant estimation and amplify it by a safety factor, which is calculated by dynamic estimation (Lera & Sergeyev, 2010).

## 7 Verification of natural perturbation

For the natural perturbed OOD score function $f(g(x_0, \delta))$, we define a new function:

$$F_{x_0}(\delta) := f(g(x_0, \delta)), \tag{14}$$

where $x_0$ is the original input. Based on Sects. 4 and 5, we have the Lipschitz continuity of $F_{x_0}$ and the Lipschitz constant $L$. The reachability problem in Definition (5) can be transformed into a multi-dimensional optimization problem:

$$\min F_{x_0}(\delta), \quad \text{subject to} \quad \delta \in [a, b]^m \tag{15}$$

### 7.1 Space-filling curve for dimension reduction

The core strategy to solve multi-dimensional optimization is transforming the $m$-dimensional parameter $\delta$ into a single dimension $d$ using a Hilbert space-filling curve (Hilbert, 1891). This enables us to then focus on the one-dimensional optimization of $d$ using Lipschitz optimization.

**Theorem 5** (*Space-filling Curve*) *A space-filling curve builds a single-valued continuous correspondence mapping the interval* $[0, 1]$ *on the x-axis onto the m dimensional hypercube* $\delta$. $p_h(d)$ *indicates the h-level approximation of the curve.* $\forall d \in [0, 1]$, *exists only one* $\delta$, *with* $\delta = (\delta_1, \delta_2, ..., \delta_m) = p_h(d)$.

With the Theorem 5, we transform the $m$ dimensional $\delta$ in to $d$ with $p_h(d)$ and define a new function $G_{x_0}(d)$:

$$G_{x_0}(d) := F_{x_0}(p_h(d)) = F_{x_0}(\delta_1, ..., \delta_m), d \in [0, 1] \tag{16}$$

The curve approximation results in a gap $B$ between the lower bounds of $G_{x_0}(d)$ and $F_{x_0}(\delta)$. Achieving $\min F_{x_0}(\delta)$ requires considering both $B$ and $\min G_{x_0}(d)$.

We search for $\min G_{x_0}$ through Lipschitz optimization. According to the Hölder condition (Finner, 1992) and construction of spacefilling curve (Lera & Sergeyev, 2010), we have the Lipschitz constant $H$ of $G_{x_0}(d)$ on $d$: $H = L\sqrt{m+3} \cdot 2^{h+1}$, with $m$ the dimension of $\delta$ and $L$ the Lipschitz constant of the original multidimensional function of $F_{x_0}$. We solve the one-dimensional Lipschitz optimization problem in Eq. (16) with Lipschitz constant $H$:

$$\min G_{x_0}(d), \quad \text{subject} \quad \text{to} \quad d \in [0, 1] \tag{17}$$

### 7.2 Lipschitz optimization with dynamic safety factor

We solve the optimization problem in Eq. (17). The optimization process is demonstrated in Fig. 3. In the first iteration in Fig. 3a, we construct zigzag lines with slopes $-H$ and $H$ passing $(0, Y_0), Y_0 = G_{x_0}(0)$ and $(0, Y_2), Y_2 = G_{x_0}(1)$, generating a cross point with value $W_0$. We project the $W_0$ to $G_{x_0}(d)$, generating $Y_1$. In the second iteration in Fig. 3b, we construct lines passing $Y_1$ and generating $\{W_0, W_1\}$. $W_0 = \min\{W_0, W_1\}$ will be chosen for the next iteration. In each iteration, we get new series of

$$W = \{W_0, W_1 ...\}, Y = \{Y_0, Y_1, ...\}$$

and we choose minimal $W$ for a new iteration. We repeat this process until $|Y_{min} - W_{min}| < \Delta$, where $\Delta$ is a predefined tolerance. The optimization process reaches convergence, generating bounds for $\min G_{x_0}(d)$:

$$W_{min} \le \min G_{x_0}(d) \le Y_{min} \tag{18}$$

During this optimization process, $W_{min} > Y_{min}$ can occur when the estimated Lipschitz constant $H$ is an understimation of the real Lipschitz constant. Inspired by Lera

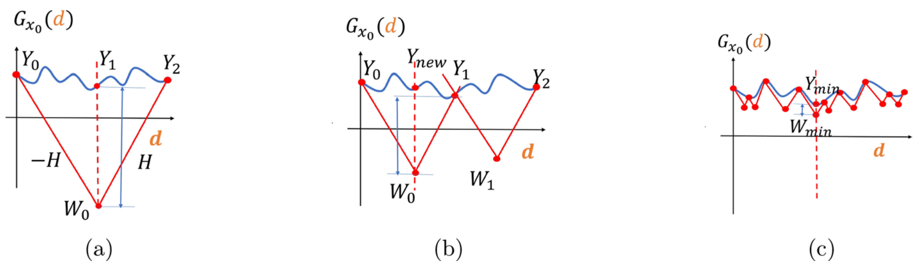



**Fig. 3** Lipschitz optimization process of the 1d function $G_{x_0}(d)$, when $H$ is an overestimation of the real Lipschitz constant, i.e., $W_{min} < Y_{min}$ always hold

and Sergeyev (2010) we adopt a dynamic estimation scheme to resolve this problem. We begin with a Lipschitz constant $H$ estimated using Extreme Value Theory (EVT) and amplify it by a factor $\beta$ gradually, where $\beta > 1$, (e.g., 1.3). The optimization iteration starts, and whenever $W_{min} > Y_{min}$ occurs, we amplify the Lipschitz constant by $\beta$. This adjustment continues until $W_{min} < Y_{min}$ is achieved, at which point we proceed with the optimization iteration.

**Algorithm 2** Lipschitz optimization with dynamic safety factor

**Input:** $L_e, m, h, \beta, G_{x_0}(d), k_{max}$
**Output:** $W_{min}, Y_{min}, \beta$

$Y_0 = G_{x_0}(0)$ , $Y_1 = G_{x_0}(1)$
$H = L_e\sqrt{m+3} \cdot 2^{h+1} \cdot \beta$
**for** $k \leftarrow 1$ to $k_{max}$ **do**
  Calculation of $Y = \{Y_0, Y_1 ... Y_k\}$
  Calculation of the new cross points $W = \{W_0, W_1 ... W_{k-1}\}$ with $H$
  $W_{min} = \min W$, $Y_{min} = \min Y$
  **for** $W_{min} > Y_{min}$ **do**
    $H \leftarrow H \cdot \beta$
    Calculate $W = \{W_0, W_1 ... W_{k-1}\}$, $W_{min} = \min W$
  **end for**
  **if** $|W_{min} - Y_{min}| \leq \Delta$ **then**
    break
  **end if**
  Choosing the interval with $W_{min}$ for the next division
  k++
**end for**
**return** $W_{min}$,$Y_{min}$, $H$

The algorithm is presented in Algorithm 2. In each iteration, we calculate the corresponding $W$ series based on the dynamic Lipschitz constant $H$. Figure 4 provides an illustration when $W_{min} > Y_{min}$ occurs in interval $j$. In Fig. 4b, we amplify Lipschitz constant by $\beta$, while $W_{min} > Y_{min}$ still holds. We further amplify the Lipschitz constant by $\beta$ until $W_{min} < Y_{min}$ in Fig. 4c. Thus, we use $H \leftarrow H \cdot \beta^2$ for the further optimization in Fig. 4.

In the whole optimization process, when the iteration number increases, with $W_{min} < Y_{min}$, we achieve $H > \lim_{d_{j+1}-d_j \to 0} |Y_{j+1} - Y_j| / |d_{j+1} - d_j| = K_{best}$. We guarantee the dynamic amplified Lipschitz constant $H$ is an overestimation of the best Lipschitz constant.

### 7.3 Lower bound in multi-dimensional space

We demonstrate how to calculate the lower bound of $F_{x_0}(\delta)$ based on the lower bound of $\min G_{x_0}(d)$.

**Theorem 6** *Let $G_{x_0}(d), d \in [0,1]$ the one dimensional function generated by Hilbert curve $p_h(d)$ and $m$ dimensional function $F_{x_0}(\delta), \delta \in [a,b]^m$. If $G_l$ is the lower bound of $G_{x_0}(d)$, then $F_l$ is a lower bound for the multi-dimensional $F_{x_0}(\delta)$, with $F_l = G_l - L \cdot D$, where $D = 2^{-(h+1)}\sqrt{m} \cdot (b-a)$, $L$ is the Lipschitz constant of $F_{x_0}(\delta)$, and $h$ is the Hilbert curve approximation level.*

***Proof*** During the construction of the Hilbert curve, the entire high-dimensional space is divided into small cubes. Within each cube, every point $\delta$ has its projected point $\delta_p$ on the Hilbert curve, representing the point on the curve $p_h(d)$ with the shortest Euclidean distance to the original point.

$$\delta_p = \mathrm{argmin}|\delta - p_h(d)|_2, \delta \in [a,b]^m, d \in [0,1] \tag{19}$$

The largest distance $D$ between $\delta$ and projected $\delta_p$ appears when the original point is located at the vertex of a subcube $D = \max|\delta - \delta_p|_2 = 2^{-(h+1)}\sqrt{m}$ with the approximation level $h$ and the $m$ dimension of $\delta$. Referring to the gap $D$ and Lipschitz continuity, we compute the discrepancy between the score of $\delta$ and its projection $\delta_p$. $|F_{x_0}(\delta) - F_{x_0}(\delta_p)| \le L|\delta - \delta_p|_2 \le L \cdot D$. Since $\delta_p$ are points on the Hilbert curve, we replace $F_{x_0}(\delta_p)$ with $G_{x_0}(d)$ according to $\delta_p = p_h(d)$ and Eq. (16), leading to $G_{x_0}(d) - L \cdot D \le F_{x_0}(\delta)$. With lower bound $G_l$ of $G_{x_0}(d)$, we have $G_l - L \cdot D \le \min G_{x_0}(d) - L \cdot D \le \min F_{x_0}(\delta)$. Thus we have proved that $F_l$ is a lower bound for $F_{x_0}(\delta)$, where $F_l = G_l - L \cdot D$ and $D = 2^{-(h+1)}\sqrt{m} \cdot (b-a)$. □

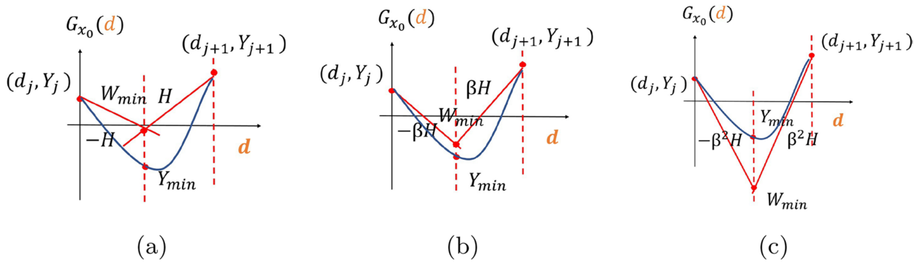



**Fig. 4** An illustration of dynamically amplifying the Lipschitz constant. **a** $H$ is an underestimation of the real Lipschitz constant, where $W_{min} > Y_{min}$. **b** Amplification by $\beta$, the new Lipschitz constant remains an underestimation. **c** amplification by $\beta^2$, $W_{min} < Y_{min}$, achieving an overestimation of the real Lipschitz constant

Based on Theorem 6 and the calculated lower bound for one dimensional $G_{x_0}(d)$ in Eq. (18), the lower bound for $F_{x_0}(\delta)$ is $W_{min} - L \cdot 2^{-(h+1)}\sqrt{m} \cdot (b-a)$. A two-dimensional example is shown in Fig. 5.

## 8 Experiments

We treat CIFAR10 and CIFAR100 as ID data respectively. Images from LSUNCrop, LSUNResize (Yu et al., 2015), Textures (Cimpoi et al., 2014), TinyImageNetCrop, and TinyImageNetResize (Deng et al., 2009) are regarded as OOD data. We adopt OOD detectors Maxsoft (Hendrycks & Gimpel, 2016), Maxlogit (Hendrycks et al., 2022), Energy (Liu et al., 2020), and KL matching (Hendrycks et al., 2022), Mahalanobis (Lee et al., 2018) and ViM (Wang et al., 2022) as benchmarks. We run our program with Pytorch on GPU V100 and 52 G RAM. We also test *Vood* on a complex black box system combing OOD detector and Yolo v4 [48]. In addition to providing the score output range, *Vood* generates perturbed samples with extreme scores, enabling evaluators to discern whether the samples are ID or OOD, as demonstrated in 7.4.

### 8.1 Illustrating example

We randomly select an ID sample for the Energy-based OOD detector. Figure 6a, b shows the distribution of $|\partial f/\partial \delta|$. For natural perturbation, *Vood* estimates the output range $[l_e, u_e]$ and compares it with the ground truth output range $[l, u]$, which is generated by grid search. For a valid verification $[l, u] \subset [l_e, u_e]$, we calculate $I = (u-l)/(u_e - l_e)$ to evaluate the accuracy as shown in Fig. 6c.

### 8.2 Comparison experiments

We assess the performance of *Vood* in comparison to baseline methods across multiple benchmarks. Given the absence of dedicated verification techniques for OOD detectors, we employ a combination of attack and classic verification approaches as our baseline methods. To achieve this estimation, we sample $N = 100$ batches with batch size $M = 50$, balancing the efficiency and estimation accuracy.

We verify the natural perturbation of rotation angle ranging [0, 30], brightness factor ranging [1, 1.5], and a combination of both. We choose DeepGo (Ruan et al., 2018) and DeepPoly (Müller et al., 2021) as baseline methods since they can provide output ranges for verification problems. The comparison of output ranges $I$ is shown in Tables 1 and 2, and the time consumption in Tables 3 and 4. The time consumption increases as the perturbation dimension grows. However, Vood primarily focuses on managing complex models with lower-dimensional perturbations. Additionally, choosing an appropriate approximation level can enhance efficiency when dealing with high-dimensional scenarios. For valid verification $I \leq 1$ always holds, and a smaller $I$ indicates looser estimation results. Notably, our method *Vood* outperforms the baselines by delivering valid and precise estimations for all benchmarks with remarkable efficiency. Deeppoly can not handle Mahalanobis or ViM OOD detectors because of their unique structure. The incorporation of a space-filling technique further enhances efficiency in high-dimensional problems. Worthy noting that

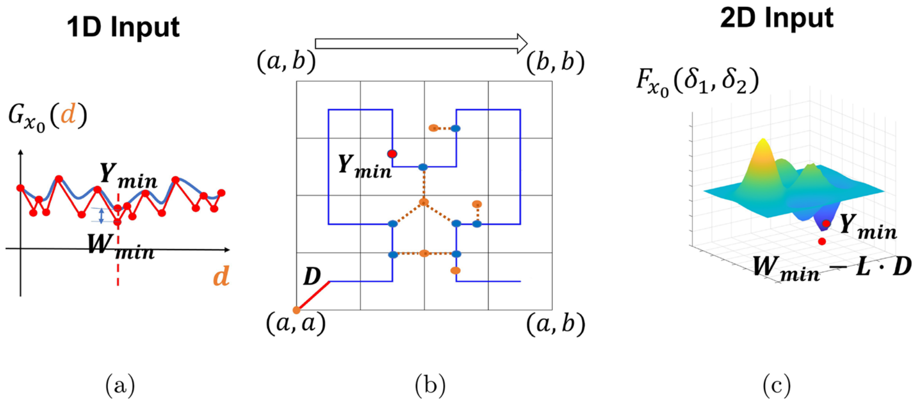



**Fig. 5** An illustrating example of estimating lower bound $F_{x_0}(\delta_1, \delta_2)$ based on the lower bound of approximated $G_{x_0}(d)$. **a** Generation bounds $W_{min} \le \min G_{x_0}(d) \le Y_{min}$ through one dimensional Lipschitz optimization **b** Samples $\delta$ (orange points) in $[a, b]^2$ projected to the points $\delta_p$ on the blue Hilbert curve. The longest distance $D$ appears when $\delta$ located at the vertex. **c** $W_{min} - L \cdot D$ is Lower bound for $F_{x_0}(\delta_1, \delta_2)$ (Color figure online)

DeepGo's utilization of dynamic Lipschitz constant estimation, devoid of assurance, can sometimes lead to invalid outcomes, shown as "X" in Tables 1, 2, 3 and 4.

## 8.3 Ablation study of parameter *M*, *N* and *h*

We study the influences of sample parameters $M$, $N$, and approximation level $h$ of the space-filling curve. Figure 7a, b and c demonstrate the relative output range $I$ to $N$, $M$ and approximation level $l$. Considering both the accuracy and efficiency, we choose $N = 100, M = 50$ in our experiments. Figure 7c shows that the output range accuracy increases with approximation level $h$ and approaches to 1. We choose $h = 10$ in high-dimensional problem for a tight estimation.

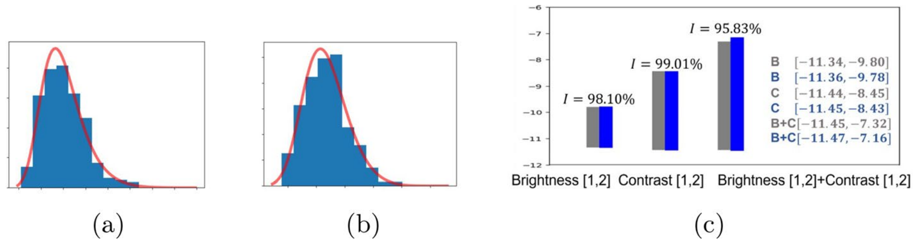



**Fig. 6** **a**, **b** Fitting the sampled $|\partial f/\partial \delta|$ to reverse Weibull distribution. **c** Output range estimation (right) for functional perturbation with brightness factor ranging [1, 2], contrast factor [1, 2] and a combination of them both. Grey bar and range stay for ground truth. Blue are *Vood* results (Color figure online)

**Table 1** $I = (u - l)/(u_e - l_e)$ for output range comparison under functional perturbation

| | | Vood (%) | Deepgo (%) | Attack (%) | Deep poly (%) |
|---|---|---|---|---|---|
| 1 | R:[0,30] | 98.88 | 98.32 | 34.53 | 42.02 |
| 2 | R:[0,30] | 99.12 | 99.56 | 43.34 | 52.42 |
| 3 | R:[0,30] | 99.2 | 99.43 | 44.27 | 45.23 |
| 4 | R:[0,30] | 98.96 | 98.01 | 43.19 | 54.12 |
| 5 | R:[0,30] | 98.73 | 98.23 | 23.12 | X |
| 6 | R:[0,30] | 98.19 | 97.82 | 38.12 | X |
| 1 | B:[1,1.5] | 99.29 | 97.25 | 34.04 | 53.23 |
| 2 | B:[1,1.5] | 99.21 | 99.43 | 48.41 | 50.43 |
| 3 | B:[1,1.5] | 98.28 | 98.23 | 34.64 | 42.68 |
| 4 | B:[1,1.5] | 99.15 | X | 39.42 | 40.18 |
| 5 | B:[1,1.5] | 97.21 | 98.05 | 39.28 | X |
| 6 | B:[1,1.5] | 98.02 | 97.85 | 31.94 | X |
| 1 | B[1,1.5], R[0,30] | 97.42 | 97.20 | 29.53 | 30.23 |
| 2 | B[1,1.5], R[0,30] | 98.02 | 97.23 | 31.40 | 40.34 |
| 3 | B[1,1.5], R[0,30] | 98.25 | 96.91 | 39.24 | 41.34 |
| 4 | B[1,1.5], R[0,30] | 97.89 | 97.49 | 34.35 | 38.23 |
| 5 | B[1,1.5], R[0,30] | 96.83 | 96.20 | 30.14 | X |
| 6 | B[1,1.5], R[0,30] | 97.10 | 96.98 | 29.18 | X |

CIFAR10 as ID samples 1. Maxsoft 2. Maxlogit 3. Energy 4. KL matching 5. Mahalanobis 6. ViM

### 8.4 Case study

We employ *Vood* on a black-box system with an OOD detector combined with Yolo V4 in [48]. The system passes the sample to an OOD detector based on a Yolo V4 model. For an ID sample, the system provides a score value and identified objects. For an OOD sample, no object can be identified. We verify a sample with a rotation range $[-6°, 6°]$. The sample is safe with a score value range $[-42, 1468, -25.0028]$, with the upper bound case at rotation 0° in Fig. 8a and lower bound case at rotation −6° in Fig. 8b. When we expand the rotation range to $[-7°, 7°]$, the system is not safe since the ID sample with −7° rotation can be identified as OOD in Fig. 8c.

**Table 2** $I = (u - l)/(u_e - l_e)$ for output range comparison under functional perturbation

| | | Vood (%) | Deepgo (%) | Attack (%) | Deep poly (%) |
|---|---|---|---|---|---|
| 1 | R:[0,30] | 97.35 | 96.13 | 28.3 | 38.2 |
| 2 | R:[0,30] | 98.74 | 96.27 | 39.12 | 41.23 |
| 3 | R:[0,30] | 99.1 | 95.42 | 25.3 | 32.8 |
| 4 | R:[0,30] | 97.63 | 94.83 | 31.42 | 32.19 |
| 5 | R:[0,30] | 98.52 | 97.29 | 35.23 | X |
| 6 | R:[0,30] | 97.82 | 96.4 | 18.23 | X |
| 1 | B:[1,1.5] | 98.15 | 96.73 | 21.5 | 29.45 |
| 2 | B:[1,1.5] | 99.61 | 96.8 | 32.34 | 35.87 |
| 3 | B:[1,1.5] | 97.18 | 97.42 | 23.4 | 41.68 |
| 4 | B:[1,1.5] | 96.49 | 95.71 | 34.12 | 40.9 |
| 5 | B:[1,1.5] | 98.81 | 96.42 | 31.23 | X |
| 6 | B:[1,1.5] | 97.5 | 96.19 | 29.11 | X |
| 1 | B[1,1.5], R[0,30] | 95.78 | 95.83 | 21.35 | 32.34 |
| 2 | B[1,1.5], R[0,30] | 96.31 | 95.28 | 24.12 | 38.63 |
| 3 | B[1,1.5], R[0,30] | 96.28 | 95.18 | 19.34 | 28.54 |
| 4 | B[1,1.5], R[0,30] | 97.4 | 96.42 | 21.31 | 31.7 |
| 5 | B[1,1.5], R[0,30] | 96.3 | 95.1 | 23.48 | X |
| 6 | B[1,1.5], R[0,30] | 97.62 | 96.23 | 19.53 | X |

CIFAR100 as ID samples 1. Maxsoft 2. Maxlogit 3. Energy 4. KL matching 5. Mahalanobis 6. ViM

**Table 3** Time consumption for natural perturbations verification

| | | Vood | Deepgo | attack | deep poly |
|---|---|---|---|---|---|
| 1 | R:[0,30] | 4.43 | 3.75 | 23.2 | 40.12 |
| 2 | R:[0,30] | 4.12 | 3.97 | 19.3 | 49.32 |
| 3 | R:[0,30] | 3.52 | 3.41 | 18.2 | 53.52 |
| 4 | R:[0,30] | 4.72 | 3.74 | 10.4 | 59.11 |
| 5 | R:[0,30] | 5.84 | 5.41 | 15.24 | X |
| 6 | R:[0,30] | 5.26 | 5.18 | 14.15 | X |
| 1 | B:[1,1.5] | 3.78 | 5.23 | 9.39 | 64.52 |
| 2 | B:[1,1.5] | 4.92 | 4.96 | 8.29 | 42.19 |
| 3 | B:[1,1.5] | 2.96 | 4.44 | 11.4 | 46.39 |
| 4 | B:[1,1.5] | 4.27 | X | 15.2 | 47.58 |
| 5 | B:[1,1.5] | 4.86 | 5.01 | 17.18 | X |
| 6 | B:[1,1.5] | 5.05 | 5.10 | 19.26 | X |
| 1 | B[1,1.5], R[0,30] | 6.79 | 12.16 | 14.39 | 151.34 |
| 2 | B[1,1.5], R[0,30] | 6.28 | 10.41 | 16.18 | 89.42 |
| 3 | B[1,1.5], R[0,30] | 7.13 | 9.12 | 18.46 | 59.19 |
| 4 | B[1,1.5], R[0,30] | 6.81 | 10.12 | 11.32 | 62.49 |
| 5 | B[1,1.5], R[0,30] | 7.14 | 9.19 | 18.67 | X |
| 6 | B[1,1.5], R[0,30] | 6.82 | 7.51 | 19.32 | X |

CIFAR10 as ID samples 1. Maxsoft 2. Maxlogit 3. Energy 4. KL matching 5. Mahalanobis 6. ViM

**Table 4** Time consumption for natural perturbations verification

| | | Vood | Deepgo | attack | deep poly |
|---|---|---|---|---|---|
| 1 | R:[0,30] | 5.12 | 7.6 | 36.6 | 50.6 |
| 2 | R:[0,30] | 4.74 | 6.5 | 25.7 | 49.8 |
| 3 | R:[0,30] | 5.9 | 7.1 | 31.5 | 45.6 |
| 4 | R:[0,30] | 6.18 | 8.4 | 21.8 | 43.7 |
| 5 | R:[0,30] | 5.8 | 7.5 | 19.8 | X |
| 6 | R:[0,30] | 6.5 | 8.7 | 30.5 | X |
| 1 | B:[1,1.5] | 7.3 | 9.3 | 19.8 | 62.19 |
| 2 | B:[1,1.5] | 5.3 | 7.5 | 21.6 | 53.8 |
| 3 | B:[1,1.5] | 6.1 | 8.6 | 19.7 | 61.5 |
| 4 | B:[1,1.5] | 6.6 | 9.5 | 23.6 | 63.6 |
| 5 | B:[1,1.5] | 6.3 | 10.1 | 25.6 | X |
| 6 | B:[1,1.5] | 7.4 | 8.6 | 21.8 | X |
| 1 | B[1,1.5], R[0,30] | 9.7 | 10.7 | 22.8 | 53.7 |
| 2 | B[1,1.5], R[0,30] | 8.31 | 11.6 | 26.8 | 62.3 |
| 3 | B[1,1.5], R[0,30] | 6.4 | 8.7 | 37.5 | 71.6 |
| 4 | B[1,1.5], R[0,30] | 7.4 | 9.4 | 25.6 | 54.9 |
| 5 | B[1,1.5], R[0,30] | 7.1 | 10.1 | 31.7 | X |
| 6 | B[1,1.5], R[0,30] | 8.7 | 9.5 | 34.3 | X |

CIFAR100 as ID samples 1. Maxsoft 2. Maxlogit 3. Energy 4. KL matching 5. Mahalanobis 6. ViM

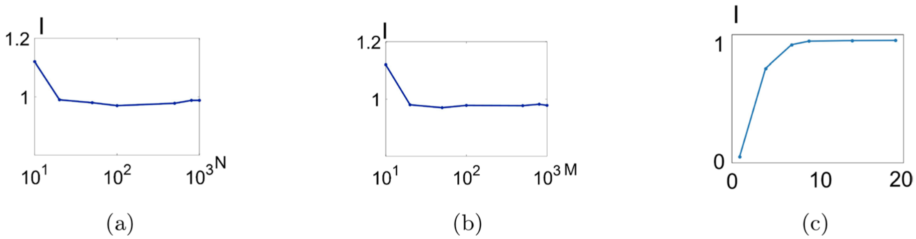
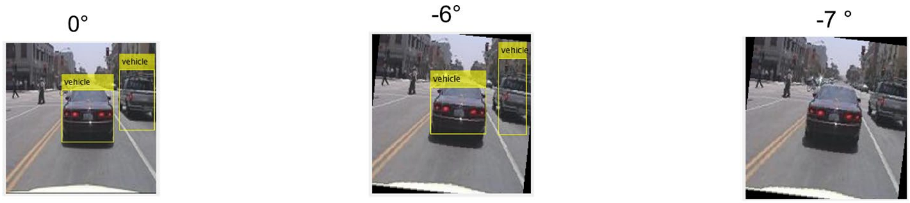



**Fig. 7** **a** M = 50, relative output range $I$ changing with $N$ **b** $N = 100$, $I$ changing with respect to $M$ **c** $I$ changing with respect to approximation level $h$




(a) -25.0028 ID (b) -42.1468 ID (c) OOD

**Fig. 8** Rotation $[-6°, 6°]$ is safe with score function upper bound **a** and lower bound **b**. Rotation $[-7°, 7°]$ is not safe with **c** identified as OOD

## 9 Conclusion

We introduce a novel verification framework named *Vood*, designed specifically to assess the reliability of OOD detectors when exposed to natural perturbations. Traditional verification tools for NNs have shown suboptimal performance or are not applicable to OOD detectors, given their distinct architecture that often deviates from standard NNs. Additionally, many OOD detectors operate as black-box systems, presenting challenges in the face of real-world perturbations. *Vood* utilizes EVT and dynamic amplified estimation of Lipschitz constant to ensure an overestimation of the Lipschitz constant. It further employs space-filling Lipschitz optimization to effectively address naturally-occurring perturbations, making it well-suited for the complexities of real-world scenarios. Our experiments highlight the exceptional performance of *Vood* in handling benchmarks and a sophisticated black-box detector based on YOLO. In future work, we aim to incorporate reachability into the training process to enhance the robustness of OOD detectors. We foresee Vood playing a crucial role in addressing safety concerns associated with intricate black-box systems, expanding its applicability to a broader range of scenarios.

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**Data Availability** No datasets were generated or analysed during the current study.

## Declarations

**Conflict of interest** The authors declare no Conflict of interest.

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